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Andrew Plaza

Staff Systems Engineer

GitHub: insipx
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SKILLS

Tools and Languages Rust, Nix/NixOS, K3S, CI/CD, C, Python, TypeScript, SQLite, Postgres, Docker, Linux, AWS, Terraform, Git
Communication English, Polish

TECHNICAL EXPERIENCE

Senior Software Engineer / Rust - Systems & Runtime Infrastructure

OCT 2023 — SEPT 2026

XMTP Labs

Remote (Philadelphia, PA)

- Core maintainer and architect of [libxmt](#) — the open-source Rust engine (MLS end-to-end encryption) powering every XMTP SDK via FFI bindings to Swift, Kotlin, Node.js (NAPI), and WASM: #1 of 62 lifetime contributors with 485 merged PRs (+288K lines) over 3 years.
- Implemented the decentralized-network (d14n) client for libxmt. Type-driven API layer, vector clock message ordering and causal ordering with out-of-order buffering.
- Designed per-PR ephemeral backend sandbox environments on ECS/Fargate for e2e feature development.
- Created [xnet](#), an ephemeral test-network orchestrator in Rust over the Docker API. Used to rehearse XMTP's centralized-to-d14n migration.
- Performance focused improvements to allow libxmt to scale to thousands of live clients per server host.
- Architected & implemented Lazy event-driven task-runner architecture for libxmt.
- Led the [WebAssembly](#) port of the full Rust MLS stack to browsers including a custom wasm persistent SQLite backend upstreamed to diesel-rs.
- Owned CI/CD and the Nix build platform for reproducible cross-compilation to iOS, Android NDK, WASM, and musl. Automated nightly releases through 8 downstream SDKs.
- Built the observability stack on the [tracing](#) crate. OpenTelemetry + Sentry support down to mobile, Datadog alerts/boards.
- Enabled iOS cross-compilation upstream in [nixpkgs](#).

Rust Blockchain Developer / Efinity

FEB 2022 — OCT 2023

Enjin

Remote (Philadelphia, PA)

- Developed substrate implementations for Fungible and Non-Fungible tokens and currencies in Efinity.
- Design and implementation of XCM (Cross-Chain Messaging) for Efinity enabling interoperability with 30 other parachains.
- Authored documentation for 30+ indie and AAA gaming partners for the features in Efinity.

Rust Software Developer / Tools - Substrate - Ethereum

NOV 2018 — MAR 2022

Parity Technologies

Remote (Berlin, DE / Philadelphia, PA)

- fully owned [Substrate Archive](#), a generic indexing tool built with Rust and PostgreSQL compatible with over 100 substrate blockchains.
- Optimized [Substrate Archive](#) to be the fastest indexer capable of indexing over 1TB of blockchain data while achieving hundred-millisecond query times.
- Created [Desub](#), a tool to decode generic bytes from a substrate transaction and correlate chain-specific types to procure human-readable transaction information.
- Trained developers for Substrate Archive and counseled external teams for its operation catered for specific use-cases.
- Presented video content educating 100+ viewers on Substrate Archive and related tooling.
- Created [dot-jaeger](#) to collect thousands of metrics per second enabling observability for parachain teams.
- Facilitated the release and review of over 5 tools such as [txwrapper](#), [jsonrpsee](#), [subxt](#), [telemetry](#).
- Implemented and drafted features in the main Substrate codebase, such as [Wasm Local-Blob Override](#), a precursor to the feature that helped core developers solve the bug which caused [Polkadot nodes/network](#) to temporarily fall offline.

PROJECTS

jupiter — Bare-Metal NixOS Fleet

github.com/insipx/jupiter

- Homelab rack (Raspberry Pi 5s + x86 nodes) in one Nix flake. k3s, custom ARM kernels, ZFS, fleet-wide deploys, and macOS remote Nix builders.

EDUCATION

Bachelor of Science in Computer Science, University of Scranton

MAY 2019

ACTIVITIES

Odyssey Momentum Hackathon - Won Conflict Prevention as Team Theriak, a group of 5 members

2020

Reached Platinum Rank in Valorant

2020

Founder ScrantonHacks

2016 – 2018